

Peace Omodiagbe Umolu | Embedded Systems Engineer | First-Class Honors Student

Shinganagh, Bansha – Co. Tipperary, Ireland

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Professional Summary

Detail-oriented Electronics & Embedded Systems Engineering student (Entering 3rd Year, First-Class Honors) with direct hands-on experience in bare-metal programming, low-level peripheral driver development, and hardware validation using ADI microcontrollers. Demonstrates exceptional discipline, maintaining First-Class academic standing while working an average of 36+ hours per week in high-responsibility healthcare and technical roles. Skilled in Embedded C, Python (PyVISA, PySerial, Pytest), I²C, SPI, and CAN communication protocols. Combines strong firmware capabilities with practical experience in high-compliance cleanroom and industrial environments.

Technical Skills

Languages: Embedded C, Python, C++, Java, Bash

Microcontrollers: ARM Cortex-M Series (Cortex-M33, Cortex-M4), EVAL-ADUCM410QSPZ, STM32, ESP32

Protocols: I²C, SPI, CAN Bus, UART, MDIO

Test Automation: Python (pytest, PyVISA, pyserial), MATLAB / Simulink

Lab Instruments: Digital Storage Oscilloscopes, Logic Analyzers (Saleae), Multimeters, Function Generators

Hardware & Tools: FreeRTOS, KiCad, Schematic Interpretation, Board Bring-up, Soldering, Git / GitHub, Linux

GUI & Host Tooling: Electron.js, React.js, JavaScript, HTML5/CSS3 (PC Diagnostic GUIs & Telemetry Tools)

Education

Technological University of the Shannon (TUS) Limerick

B.Eng. (Honors) in Electronic Engineering with Computer Systems, First-Class Honors 2024–Present

Achieved while managing 36+ hours/week in employment. Coursework: Microcontroller Systems, Embedded C Programming, Digital Signal Processing, Analog & Digital Circuit Design, Data Communications, PCB Design & EDA.

IES Cabanyal Valencia, Spain

Advanced Certificate / Higher Certificate (NFQ Level 6) – Electrical & Electronics, Distinction 2013–2015

Focused on industrial electrical installations, PLC programming, electronic component troubleshooting, and schematics.

Technical Projects

ADI EVAL-ADUCM410QSPZ Personal Portfolio

Smart Single-Cell Battery Management System (BMS) & HIL Test Rig 2025–2026

- Firmware Development (Embedded C): Engineered a multi-threaded FreeRTOS application monitoring simulated battery parameters (V_{cell} , I_{load} , temperature). Configured the 16-bit ADC with DMA circular buffering for continuous background sampling without CPU overhead.
- Communications & Algorithms: Implemented non-blocking I²C and SPI telemetry drivers alongside a real-time Coulomb Counting State of Charge (SoC) estimation algorithm calibrated against OCV lookup tables.
- Python Test Automation: Built an automated PyVISA test suite using **pytest** and **pyserial** to sweep bench supply voltages, log serial telemetry, and validate over-voltage fault trip execution within 50 ms.

Professional Experience

Acquired Brain Injury (ABI) Ireland Tipperary, Ireland

Rehabilitation Healthcare Assistant July 2026–Present

Deliver specialized person-centered rehabilitation support for clients with acquired brain injuries, maintaining rigorous health, safety, and compliance documentation under strict protocols.

Access Healthcare Ireland

Healthcare Assistant (Agency) 2024–Present

Provide high-demand care support across healthcare facilities, managing an average of 36 hours weekly alongside full-time engineering studies.

Abbott Vascular	Clonmel, Ireland
Technical Operative / Assembler	March 2025–Present
Execute precise assembly and inspection of medical devices in a high-compliance cleanroom environment, strictly adhering to Standard Operating Procedures (SOPs) and quality standards.	
Inforlandia	Valencia, Spain
IT Support Technician & Frontend Developer	2017–2021
Diagnosed and resolved hardware, board-level, and network faults across client systems. Configured software environments, drivers, and user interfaces.	
B-Market / Jobelsa	Spain
Electronic Technician & PLC Programmer	2015–2017
Assembled and calibrated electronic tools. Programmed and configured PLCs for industrial production lines using electrical schematics.	

Additional Information

Languages: Fluent in English and Spanish

Certifications: PHECC First Aid, Manual & Patient Handling, Safeguarding Adults, AMRIC Infection Control

References: Available upon request